

Joint Research Proposal (Public Version)

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Submitted to: Daiichi Sankyo Co., Ltd. — Open Innovation Office

About this document. This is the public version of a joint research proposal that Fox2E Studio Inc. has submitted to the Open Innovation Office of Daiichi Sankyo Co., Ltd. In the interest of public benefit and transparency, we are publishing it on our corporate website.

Executive Summary (for general readers)

This document is a joint research proposal addressed to the Open Innovation Office of Daiichi Sankyo Co., Ltd. It is a public-interest document that places patient safety and public health value above all other considerations, and it is published on our corporate website to ensure transparency.

DXd-based antibody-drug conjugates (ADCs), including T-DXd, are widely used as breakthrough cancer therapies. At the same time, drug-related interstitial lung disease (ILD) / pneumonitis has been reported as a clinically significant adverse event: a pooled analysis of nine T-DXd monotherapy studies (n=1,150) reported an ILD/pneumonitis incidence of 15.4% and grade 5 events of 2.2% (Powell et al. 2022, ESMO Open 7(4):100554). Current patient-selection methods are based primarily on antigen expression, which means that patients with limited expected benefit may still be exposed to the risk of serious adverse events. A more precise framework for evaluating the risk-benefit balance is therefore warranted.

As a complementary approach to this challenge, we are proposing the joint validation and clinical-implementation feasibility assessment of a diagnostic technology that we refer to as a "Payload CDx" (payload companion diagnostic), within the framework of a joint research collaboration with Daiichi Sankyo. Because this diagnostic is built on top of an existing, validated pathology assay rather than a brand-new platform, we believe its feasibility for validation is comparatively high.

The proposer is a former Daiichi Sankyo employee (2007–2018) and has worked in this field for more than ten years. This proposal does not contain criticism of any specific company, nor does it serve any private interest of the proposer. It is structured so that any stakeholder — patients, clinicians, regulators, shareholders — can independently verify its content. Technical terms used in this document are explained in Section 13 (Glossary).

1. Title of the Proposal

Proposal for the development of a Payload Companion Diagnostic (P-topol-Dx) for DXd-payload antibody-drug conjugates (ADCs).

Spirit of the proposal. This proposal is grounded in a public-interest objective: prioritising the lives and safety of cancer patients above all else. Through early clinical implementation of a Payload CDx, we aim to help create public-health value — maximising response rates with DXd-ADC therapy, optimising the risk-benefit balance for adverse events, allocating healthcare resources more efficiently, and making the contribution of DXd-ADC therapy to the broader healthcare system more visible. Regardless of whether this proposal is accepted, the proposer will continue to engage academically with the public-interest discussion around patient safety and resource optimisation in this field.

2. Purpose of the Proposal

The principal purpose of this proposal is to improve the precision with which DXd-ADC therapy is administered — in particular, to reduce administration to patients who are unlikely to respond — through early clinical implementation of a Payload CDx. The intended outcome is a reduction in avoidable adverse events and more efficient allocation of healthcare resources. The proposer places top priority on maximising patient safety and public-health value, and will continue academic work in this field for the public good regardless of the outcome of this proposal.

This document follows a standard format for transparent, accountable public-interest communication in the pharmaceutical research domain, and it is written so that third parties — the medical community, regulators, patient organisations — can verify and discuss it independently. The scientific basis of the proposal rests on peer-reviewed literature and public reports indexed in PubMed and KAKEN, so the underlying evidence is openly accessible for independent review.

3. Background

The proposer (Yusuke Yokouchi) worked on ADC development, translational research, and CDx within Daiichi Sankyo. Areas of professional expertise include veterinary medicine (licensed veterinarian), pathology (peer-reviewed publications), ADCs (hands-on experience visualising *in vivo* MOA), translational research (peer-reviewed publications produced jointly with pathology departments at designated cancer centres), CDx (in or around 2015 the proposer came across the KAKEN report by Ando et al. on topol-pS10 in a public database, became interested in its clinical applicability, and has tracked the field since), and AI.

Daiichi Sankyo's flagship pipeline — the DXd-ADC platform (Enhertu® / trastuzumab deruxtecan, Datroway® / datopotamab deruxtecan, HER3-DXd, and others) — is a third-generation ADC family that uses DXd, an exatecan-derived topoisomerase I (topoI) inhibitor, as its payload. These drugs have shown striking clinical efficacy across multiple tumour types, and their indications continue to expand.

Patient selection today, however, relies on Antigen CDx — measurement of target-antigen expression (HER2, TROP2, HER3, and so on). Sensitivity prediction based on the mechanism of action of the payload

itself — what we call a Payload CDx — has not yet been implemented. As a result, even patients whose tumours express the target antigen may receive treatment despite low intrinsic sensitivity to DXd, which can dilute response rates and create economic inefficiency.

In recent years, Ando et al. (J Histochem Cytochem 2018, PMID: 29587004; Clin Colorectal Cancer 2021, PMID: 33731288) developed an IHC-based biomarker, P-topol-Dx (a topol-pS10 detection assay), that predicts sensitivity to topol inhibitors. In a retrospective multi-cohort validation (n=282; eight institutions in four countries), they reported the following performance:

- Colorectal cancer (n=176, threshold 40%): sensitivity 87.5%, specificity 70.0%, NPV 87.0%
- Gastric cancer validation set (n=27, threshold 35%): sensitivity 82.4%, specificity 70.0%

What is particularly notable is that, in the Discussion of Ando et al. (2021), Enhertu (a DXd-ADC) is named explicitly as a candidate target drug for P-topol-Dx ("2 third-generation ADCs (SN38 in Trodelvy and Exatecan in Enhertu) are topol inhibitors."). In other words, the scientific rationale for applying a Payload CDx to one of Daiichi Sankyo's flagship assets has already been laid out, in print, in independent third-party academic work.

4. Scientific Basis of the Proposal

Mechanistic rationale for P-topol-Dx

According to Ando et al. (2017), the cellular response to camptothecins (CPTs) — a class that includes exatecan and DXd — is determined along the following pathway:

- Formation of the topol–DNA–CPT cleavable complex
- Replication-fork collapse, generating DNA double-strand breaks (DSBs)
- Phosphorylation of topol at Ser10 by DNA-PKcs (topol-pS10)
- K48-linked ubiquitination by BRCA1/BARD1
- Rapid degradation of topol via the ubiquitin-proteasome pathway, leading to resistance

Tumour cells with high topol-pS10 expression degrade topol rapidly and therefore tend to be resistant to CPTs. Tumour cells with low topol-pS10 expression, by contrast, retain the cleavable complex for longer, accumulate DSBs, and are sensitive. The same logic applies, in principle, to ADCs that carry a DXd payload.

Evidence of topol inhibition in DXd-ADCs

In a preclinical pharmacology study published by Daiichi Sankyo itself (Ogitani et al. 2016, Clin Cancer Res 22(20):5097–5108, PMID: 27026201), DS-8201a (trastuzumab deruxtecan / Enhertu) was shown to exert HER2-dependent antiproliferative activity and in vivo antitumour activity, and to do so via topol inhibition by DXd released after linker cleavage. Even within Daiichi Sankyo's own published literature, then, there

is strong support for the view that topol inhibition is the principal cell-death pathway for DXd-ADCs and that topol-pS10 should function as a sensitivity-prediction marker.

Mechanism and clinical incidence of DXd-ADC-related ILD

On the mechanism of DXd-ADC-related ILD/pneumonitis, Kumagai et al. (2020, Cancer Sci 111(12):4636–4645, PMID: 33051938) reported that target-independent uptake of T-DXd by alveolar macrophages may be involved. In monkey lung, HER2 expression was confined to the bronchial level, whereas the ILD lesions appeared at the alveolar level; in addition, free DXd administered at equivalent or higher exposure did not produce lung toxicity. The authors therefore inferred that T-DXd is first taken up by alveolar macrophages, undergoes cathepsin-B-mediated linker cleavage inside those cells, and then triggers ILD via bystander diffusion of DXd into surrounding tissue.

The same paper also describes HER2 expression and DXd-positive findings in regenerative alveolar epithelium (cells showing regenerative changes such as type II alveolar epithelial cells) within the ILD lesions. This suggests a secondary aggravation pathway: during the tissue-regeneration phase that follows initial injury, T-DXd may directly target the alveolar epithelium via HER2 expressed on regenerative cells. In other words, the onset of ILD (Phase 1 — a HER2-independent initial trigger) and its progression (Phase 2 — HER2-dependent secondary aggravation) may be driven by distinct mechanisms. Suppressing the initial trigger should therefore also reduce the chance of progression into the secondary pathway.

On clinical incidence, Powell et al. (2022, ESMO Open 7(4):100554, PMID: 35963179) — a pooled analysis of nine T-DXd studies (n=1,150) — reported drug-related ILD/pneumonitis at 15.4%, grade ≥ 3 severe events at 3.5%, and grade 5 (fatal) events at 2.2%. Under CTCAE v5.0, grade 3 ILD/pneumonitis is defined as an event requiring medical intervention such as supplemental oxygen, and the monkey study by Kumagai et al. (2020) documented that some findings — alveolar macrophage aggregation, interstitial inflammation — persisted into the recovery period. There is therefore reasonable concern that even surviving patients may experience lasting effects on respiratory function.

This mechanism (HER2-independent initial trigger plus HER2-dependent secondary aggravation) is independent of the tumour-response prediction pathway (sensitivity assessment via topol-pS10). Patient selection by Payload CDx reduces unnecessary DXd-ADC exposure in patients who are unlikely to respond, which in turn reduces the opportunity for serious adverse events to occur. By dampening the Phase 1 initial trigger in particular, it should also reduce the chance of progression into the Phase 2 secondary pathway. We therefore position the Payload CDx as a complementary approach that contributes to the public-interest goal of reconciling patient safety and healthcare-resource optimisation.

Expansion of the HER2 spectrum, the move toward all-comer eligibility, and why a Payload CDx matters strategically

Recent work re-classifying the HER2 expression spectrum has shown that the population eligible for T-DXd (HER2-positive, HER2-low, and HER2-ultralow) covers the great majority of breast cancer patients. Lv et al. (2025, Breast Cancer Res 27(1):45, PMID: 40134013), a multicentre study of 2,869 patients across ten Chinese institutions, reported HER2-positive at 21.7%, HER2-low at 54.5%, HER2-ultralow at 10.6% (combined 86.8%), and HER2-null (true negative) at 13.2%. Morrar et al. (2025, Mod Pathol 38(12):100901, PMID: 41033579), a re-classification study at Memorial Sloan Kettering Cancer Center (n=400), reached similar numbers: HER2-positive 8%, HER2-low 47%, HER2-ultralow 29% (combined 84%), HER2-null 16%.

Antigen CDx based on HER2 IHC therefore remains an indispensable basis for patient selection. At the same time, as eligibility expands to include HER2-low and HER2-ultralow patients, a complementary stratification axis — one that further refines the prediction of clinical benefit — is becoming increasingly important.

When this move toward near-all-comer eligibility is combined with the fact that ILD onset includes a HER2-independent initial pathway (target-independent uptake by alveolar macrophages, Kumagai et al. 2020), a complementary patient-stratification axis based on payload mechanism — distinct from antigen expression — becomes strategically more, not less, necessary if precision in patient selection is to be preserved while serious-adverse-event risk is managed. The Payload CDx is precisely this complementary axis: used alongside an Antigen CDx, it functions as a diagnostic tool that helps reconcile patient safety with maximal clinical value during this era of expanding eligibility.

Status of P-topol-Dx assay development

Standardisation of the IHC assay has already been completed by Ando et al. (2018):

- Mouse monoclonal antibody, clone 357.3.1C1.H5.H7 (IgG1κ)
- Cross-platform validation completed across two platforms, Biocare Intellipath and Ventana Benchmark Ultra
- All specificity controls established (isotype-matched IgG, peptide competition, calf intestinal phosphatase treatment)
- A %PN quantification protocol established on Aperio AT2 with the V9 nuclear algorithm
- Reproducibility confirmed across multiple operators and multiple days

In other words, this proposal is not about developing a new assay. It is about clinically implementing an already-validated IHC assay as a Payload CDx for the Daiichi Sankyo DXd-ADC family.

Why now: maturation of independent validation data, 2018–2025

As mentioned, the proposer first encountered the KAKEN report by Ando et al. on topol-pS10 around 2015 in a public database, became interested in its clinical applicability, and has continued to follow the

accumulation of independent validation data ever since. At that time, the scientific foundation of the field was less mature than it is today, and the moment for a formal proposal had not yet arrived. Between 2018 and 2025, however, independent validation data accumulated to the point where the scientific basis for this proposal is now, in our view, well established. The specific milestones are as follows:

2014. Ando K, "Search for novel biomarkers of Topol-inhibitor sensitivity in colorectal cancer and development of new therapies" (KAKEN report, project number 26861082). This report set out the direction of biomarker discovery focused on the topol-degradation mechanism and suggested that topol-pS10 may correlate with CPT-11 sensitivity. It also reported the generation of a topol-pS10 monoclonal antibody and confirmation of staining on resected colorectal cancer specimens by IHC.

2017. Ando et al. (Oncotarget, PMID: 28415827) elucidated, at the molecular level, the CPT-resistance mechanism centred on topol-pS10, as described above.

2018. Ando et al. (J Histochem Cytochem, PMID: 29587004) established a standard operating procedure for the P-topol-Dx IHC assay and completed validation across multiple platforms, as described above.

2021. Ando et al. (Clin Colorectal Cancer, PMID: 33731288) completed retrospective multi-cohort validation in 282 cases across eight institutions in four countries, reporting sensitivity 87.5% and NPV 87.0% in CRC, as described above.

2025. Ando et al. (ASCO 2025, JCO 2025;43(4_suppl):217) reported that combination with a proteasome inhibitor suppresses topol degradation and enhances irinotecan efficacy in resistant CRC cells and a mouse model.

In parallel, since 2020 the clinical success and indication expansion of DXd-ADCs such as Enhertu and Datroway — including the move of Enhertu into first-line treatment and into HER2-ultralow populations — has steadily increased the number of eligible patients and, with it, the magnitude of the impact on healthcare resources and on national health-insurance finances. The clinical and economic upside of stratifying these patients via a Payload CDx has therefore expanded along the same trajectory, which is what makes the timing of this proposal more compelling than it would have been even a few years ago. Taken together, the accumulated evidence and the trajectory of indication expansion provide a sound scientific and practical basis for revisiting this proposal now.

Therapeutic option expansion: the rescue-strategy angle

The 2025 ASCO findings from Ando et al. — that combination with a proteasome inhibitor suppresses topol degradation and enhances irinotecan efficacy — point to a way of expanding the clinical value of a Payload CDx beyond stratification (deciding whether to administer) to rescue-strategy development (expanding the menu of therapeutic options). Specifically:

- **Rescue therapy for Topol-pS10-positive cases:** A treatment option in which a proteasome inhibitor is combined to suppress topol degradation and restore DXd-ADC sensitivity is, in principle, achievable.
- **Establishing a second-line regimen via re-biopsy after DXd-ADC resistance emerges:** In cases where DXd-ADC resistance becomes evident after treatment has begun, if topol-pS10 positivity is confirmed at the time of re-biopsy, a diagnostic algorithm for shifting to a proteasome-inhibitor combination regimen can be constructed. This would offer continued therapeutic options to patients in whom resistance has emerged.
- A new development path — DXd-ADC plus proteasome inhibitor as a combination strategy — opens up, with the Payload CDx serving as a co-developed companion biomarker.
- On this view, the proposal is not merely the implementation of a predictive marker. It can also serve as the starting point for the development of a new treatment regimen.

5. Proposed Joint Research

Phase 1 — Feasibility verification

- P-topol-Dx staining on Daiichi Sankyo's preclinical DXd-ADC samples (xenograft- and PDX-model tissues)
- Correlation analysis between topol-pS10 levels and in vivo antitumour efficacy
- Temporal and spatial correlation analysis with γ H2AX upregulation

Phase 2 — Validation in clinical specimens

- P-topol-Dx evaluation of archived clinical-trial specimens for Enhertu, Datroway, and so on (responders and non-responders)
- Construction of a hierarchical predictive model combining the existing Antigen CDx (e.g. HER2 IHC)
- Identification of the antigen-positive but P-topol-Dx-negative population, and verification of its clinical significance

Phase 3 — CDx co-development

- CDx co-development including engagement with regulators (PMDA, FDA)
- Clinical implementation as a Daiichi Sankyo-led Payload CDx

6. Areas Where the Proposer Can Contribute

ADC MOA visualisation

In the course of work in the ADC research field, the proposer built and operated a technique for visualising the mechanism of action of an ADC in mouse xenograft studies, using only standard IHC equipment, before the PID technology was commercialised. Specifically, on tumour samples collected at multiple time points after ADC administration, a method of overlaying DAB-converted serial IHC sections was applied. This made it possible to observe the entire pharmacological cascade — ADC binding to tumour cells, internalisation (with concomitant decrease in target-antigen expression), DNA-damage response (γ H2AX upregulation), and tumour-cell death — at single-cell to subcellular resolution, with a temporal axis.

Suzuki et al. (2021, Clin Cancer Res 27(14):3970–3979, PMID: 33980613) later used PID (phosphor-integrated dots) to visualise the bystander effect of a HER2-ADC (Enhertu / T-DXd). The proposer's approach is complementary to Suzuki 2021: it differs in detection technology (DAB vs. PID) and in the axis being observed (intracellular pharmacological cascade vs. spatial bystander diffusion). It also includes content that predates the related virtual multiplexing work reported by Harms et al. (2023, PMID: 37105494). MOA visualisation across the full DXd-ADC platform — together with an integrated biomarker strategy that includes P-topol-Dx — has the potential to lift the value of the entire Daiichi Sankyo ADC platform, and we propose that it be used within a joint research framework.

7. Expected Outcomes and Public-Health Value

This joint research treats improvement of patient safety as the highest-order public-interest objective and offers a framework for co-creating, with Daiichi Sankyo, the clinical, economic, and strategic value described below. Each item in this section can also be read as a public-health benefit, and together they constitute the concrete substance of the proposal's public-interest scope.

7.1 Patient and clinical value

- **Improved response rate.** Maximising the therapeutic effect of DXd-ADCs by improving the precision with which patients with high topol-inhibitor sensitivity are selected — in other words, more accurately identifying the responder population.
- **Optimal management of ILD risk.** Concentrating treatment on the population most likely to benefit, thereby optimising the risk-benefit balance between efficacy and adverse events — particularly ILD, the principal serious adverse event of DXd-ADCs.
- **Better-informed consent.** A Payload CDx is not currently in clinical use, so sensitivity-prediction information based on payload mechanism cannot be incorporated into the physician–patient decision-making conversation. This is not a problem specific to any single company; it is a structural gap in the clinical implementation of diagnostic technology. Implementing a Payload

CDx narrows that gap and substantively expands the information available to patients when choosing therapy — a public-interest benefit in itself.

7.2 Economic and system value

- **More appropriate use of healthcare expenditure.** Avoiding administration to non-responders and concentrating limited high-cost resources — drug, infusion management, adverse-event monitoring — on the patient population most likely to benefit; in short, optimal allocation of healthcare resources.
- **Visibility of the contribution to the healthcare system.** Increasing the social value of the DXd-ADC platform that Daiichi Sankyo holds and strengthening the explanatory power of that contribution for stakeholders — regulators, payers, patient organisations.

7.3 Strategic and scientific value

- **Competitive advantage.** Erecting a barrier to entry in the third-generation ADC market by being first to implement a Payload CDx clinically.
- **Scientific value.** Establishing a sensitivity-prediction paradigm grounded in the mechanism of action of the ADC.
- **Value for new therapeutic development.** Serving as the starting point for rescue-strategy development based on combination with a proteasome inhibitor.
- **Brand and credibility value through being first in the world to clinically implement a Payload CDx.** The first clinical implementation, anywhere in the world, of a diagnostic that quantitatively predicts therapeutic effect from the mechanism of action of the payload would be a strategic asset that symbolises Daiichi Sankyo's global competitiveness and R&D capability, contributing to trust from regulators, clinicians, and patient organisations and to brand value.
- **Complementary contribution to the development of ADC MOA visualisation.** Visualisation of the spatial intratumoural dynamics of DXd-ADCs (the bystander effect) has already been established within Daiichi Sankyo through prior work (Suzuki et al. 2021, PMID: 33980613). The proposer's independent technique — visualisation of the intracellular pharmacological cascade of an ADC — is positioned as a unique, complementary contribution at a different (subcellular) resolution. Because it can be carried out using only standard IHC equipment, it also has practical advantages for clinical translation.

8. Distinctive Features of This Proposal

What distinguishes this proposal from a typical external proposal flows from the proposer's background and long-term familiarity with the field:

- The proposer has hands-on experience in ADC and CDx work and understands the R&D-level questions in this domain.
- The proposer has more than ten years of continuous engagement with the field, dating back to the 2014 KAKEN report, and has consistently observed the accumulation of independent validation data by Ando et al. (2017, 2018, 2021, 2025).

9. A Request Regarding Internal Consideration

This proposal addresses a topic of high public-interest relevance: optimising the risk-benefit balance for serious adverse events — especially interstitial lung disease — associated with DXd-ADCs. It also touches on questions that warrant management-level attention, such as the overall strategic direction of the DXd-ADC platform and the cultivation of trusting relationships with regulators, clinicians, and patient organisations. We would therefore be grateful if it were considered not only by the relevant R&D, clinical-development, safety, and regulatory functions, but, where appropriate, also at the executive level.

10. Declaration of Conflicts of Interest

The proposer (Yusuke Yokouchi, Representative Director, Fox2E Studio Inc.) hereby declares that none of the following conflicts of interest apply to this proposal:

- Personal or financial relationships with Kyushu University or its affiliated organisations: none.
- Personal or professional relationships with Dr. Koji Ando (lead author of the papers cited in this proposal): none.
- Relationships with the Dana-Farber Cancer Institute or its affiliated organisations: none.
- Relationships with Boston University or its affiliated organisations: none.
- Commercial interest in the commercialisation of P-topol-Dx: none.
- Sponsorship, contracts, or benefits received from competing diagnostic or pharmaceutical companies: none.
- Any monetary benefit contingent on the acceptance of this proposal: none.
- Honoraria or commissioned research funds from authors of the cited papers or from related research institutions: none.

The proposer states that the motivation for this proposal lies in the public-interest objectives of improving patient safety and allocating healthcare resources efficiently, and that it does not arise out of any interest-based relationship with any specific researcher, company, or organisation.

Through training received during his time at the Daiichi Sankyo Safety Research Laboratories, the proposer learned two non-negotiable principles: that patient safety is the highest priority, and that important

records and proposals should be preserved in a form that is authentic and auditable. Through subsequent practical work in the Biomarker Promotion Department, he also learned how important it is to publicise meaningful research output and proposals to the academic and medical communities in an appropriate form, so that the knowledge does not dissipate and instead becomes an information asset that supports follow-on research and clinical implementation. This proposal is presented in line with those principles, and, in the spirit of public benefit, has been published on our corporate website. The published version differs in granularity from the version formally submitted to the addressee.

11. Contact

We would be grateful to receive an update on the status of your consideration of this proposal by email.

Proposer:

Yusuke Yokouchi, Representative Director, Fox2E Studio Inc.

e-mail: omitted in this public version

Note: an English-language version of this proposal is also available. Please request it as needed for review by overseas teams.

12. References

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13. Glossary (for general readers)

The principal technical terms used in this document are explained below by category.

Drugs and therapeutic technologies

- **ADC (antibody-drug conjugate).** A next-generation anti-cancer drug in which a cytotoxic agent (the payload) is attached to an antibody. It selectively attacks cancer cells expressing the target antigen.
- **DXd.** A payload component developed in-house by Daiichi Sankyo. An exatecan derivative with topoisomerase I inhibitor activity.
- **DXd-ADC.** Collective term for ADCs that use DXd as payload (Enhertu, Datroway, etc.).
- **Enhertu® (trastuzumab deruxtecan, T-DXd, DS-8201a).** A HER2-targeted DXd-ADC developed and marketed by Daiichi Sankyo.
- **Datroway® (datopotamab deruxtecan).** A TROP2-targeted DXd-ADC developed and marketed by Daiichi Sankyo.

- **Camptothecins (CPTs).** A family of topoisomerase I inhibitors that includes irinotecan, topotecan, and others. DXd belongs to the same class.

Targets and antigens

- **HER2, TROP2, HER3.** Target proteins expressed on the surface of cancer cells. The antibody portion of an ADC recognises these molecules.
- **HER2-positive / HER2-low / HER2-ultralow / HER2-null.** A four-tier classification of breast cancer based on the level of HER2 expression (positive, low, ultralow, negative).
- **topol (topoisomerase I).** An enzyme that relieves DNA torsion during replication and transcription. The target of CPT-class drugs.
- **topol-pS10.** topol phosphorylated at serine 10. The basis of the sensitivity-prediction marker discussed in this proposal.

Diagnostic technologies (CDx)

- **CDx (companion diagnostic).** A diagnostic technology used to decide whether — and how — to use a particular therapeutic agent.
- **Antigen CDx.** A CDx that measures the level of expression of a target antigen. Example: HER2 IHC.
- **Payload CDx.** A CDx that predicts cellular sensitivity to a payload. The concept addressed by this proposal.
- **P-topol-Dx.** An IHC assay that detects topol-pS10. The core technology of this proposal.

Pathology and assay technology

- **IHC (immunohistochemistry).** A standard pathology technique that uses antibodies to detect specific proteins in tissue.
- **FFPE.** Formalin-fixed paraffin-embedded — the global standard tissue-preservation format.
- **Xenograft.** An animal model in which human-derived tumour tissue is implanted in a mouse.
- **PID (Phosphor-Integrated Dots).** A high-brightness fluorescent nanoparticle imaging technology developed by Konica Minolta.

Adverse events

- **ILD (interstitial lung disease).** A serious adverse event in which lung tissue becomes inflamed and fibrotic. The principal adverse event of DXd-ADCs.

- **Bystander effect.** A phenomenon in which payload released from a target cell also acts on neighbouring non-target cells.
- **CTCAE.** Common Terminology Criteria for Adverse Events, the severity-grading system maintained by the U.S. National Cancer Institute. Grades range from 1 to 5 (5 = death).
- **Grades 3–5.** CTCAE severity tiers. Grade 3 = severe, Grade 4 = life-threatening, Grade 5 = death.
- **SAE (Serious Adverse Event).** An adverse event with serious consequences such as death or prolonged hospitalisation.

Mechanisms and molecules

- **DSB (DNA double-strand break).** A lesion in which both DNA strands break simultaneously. If unrepaired, it leads to cell death.
- **Cathepsin B.** A lysosomal protease. Involved in cleavage of an ADC linker.
- **Alveolar macrophage.** A phagocytic immune cell that resides in the alveoli.
- **Type II alveolar epithelial cell.** A type of epithelial cell that constitutes the alveolar wall and plays a central role in alveolar regeneration after injury.
- **Ubiquitin Proteasome Pathway (UPP).** The intracellular protein-degradation system. Ubiquitinated proteins are degraded by the proteasome.
- **Proteasome inhibitor.** A drug that inhibits proteasome function. Bortezomib and others are used clinically.

Statistics and diagnostic metrics

- **Sensitivity.** The proportion of people with the condition who are correctly identified as positive.
- **Specificity.** The proportion of people without the condition who are correctly identified as negative.
- **NPV (negative predictive value).** Among those whom the test classifies as negative, the proportion who actually do not have the condition.
- **PPV (positive predictive value).** Among those whom the test classifies as positive, the proportion who actually have the condition.

Regulators and databases

- **PMDA.** Pharmaceuticals and Medical Devices Agency — the Japanese drug regulator.
- **FDA.** U.S. Food and Drug Administration — the U.S. drug regulator.

- **PubMed / PMID.** The article database of the U.S. National Library of Medicine and the unique identifier for an article in that database.
- **DOI.** Digital Object Identifier — a persistent identifier for academic publications.
- **KAKEN.** The Japanese database of research outputs from Grants-in-Aid for Scientific Research.
- **ASCO.** American Society of Clinical Oncology.
- **JCO.** Journal of Clinical Oncology — a leading journal in clinical oncology.

14. Disclaimer

This document is an academic, public-interest proposal based on publicly available literature. It is not intended to defame any specific drug, company, or individual. It does not constitute medical advice, advertising of any drug or diagnostic, investment solicitation, or a guarantee of the efficacy of any unapproved diagnostic technology. Actual treatment decisions must be made on the basis of the judgement of physicians and other healthcare professionals and on the approvals and guidance of national regulators.